

PROJECT REPORT

Web Scraping, Data Wrangling & Visual Storytelling

Website: flip.kz (Books Catalog)

Domain: E-Commerce & Retail - Books

Team members: Lailya Mukhambetova, Dana Beisbayeva, Tomiris Bakyt, Aizere Dauletbayeva

1. INTRODUCTION & MOTIVATION

flip.kz is one of Kazakhstan's largest e-commerce platforms for books and media, hosting thousands of titles across genres including Science Fiction, Romance, Action, Manga, and Biography. Book market has received very little public data analysis, making flip.kz an ideal candidate for original research.

Our team chose this domain because we were genuinely curious about how books are priced in Kazakhstan, whether discounts are strategically applied across genres, and whether community ratings actually correlate with commercial value. We also wanted to understand which publishers dominate the market and what makes a book 'popular' in this ecosystem.

2. DATA SOURCE DESCRIPTION

The data was scraped from the flip.kz book catalog using paginated subsection URLs of the form: <https://www.flip.kz/catalog?subsection={sub}&page={page}>

Five subsections (genres) were targeted: Фантастика (Sci-Fi), Романы (Romance), Боевики (Action), Манга и комиксы (Manga), and Биография (Biography). The final raw dataset contained 7,618 records across 11 fields

	Field Name	Type	Example
1	title	string	Гордость и предубеждение
2	price	string → Int64	3 735 ₸ → 3735
3	discount	string → float	-20% → 20.0

4	bonus	integer	37
5	cover	string	Мягкая обложка
6	year	integer	2023
7	publisher	integer	2023
8	rating	float	4.8
9	reviews	string → int	34 отзыва → 34
10	stock	string → date	На складе → 2026-04-27
11	section	string	Фантастика

3. SCRAPING METHODOLOGY

- Tool Choice: BeautifulSoup4 + Requests. We used BeautifulSoup4 combined with the Requests library. flip.kz renders book listing pages as static HTML. This approach is faster, lighter, and easier. BeautifulSoup was the simplest and most efficient choice for parsing HTML responses.
- The catalog follows a predictable paginated URL pattern:
`https://www.flip.kz/catalog?subsection={id}&page={n}` We iterated over 5 subsection IDs (134 = Sci-Fi, 142 = Romance, 135 = Action, 970 = Manga, 158 = Biography) and incremented the page number from 1 until a page returned no book cards — this was our stop condition.
- Every individual field extraction was wrapped in a try/except block. If a tag was missing on a card (e.g. no rating for an unreviewed book), that field was set to None instead of crashing the entire script. We also added request timeouts and a retry mechanism for failed HTTP responses.
- Before scraping, we checked [flip.kz](https://www.flip.kz) and confirmed that the catalog is not disallowed. We added a 1.2-second delay between every request to avoid overloading the server. We identified ourselves with a descriptive User-Agent header: *"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0"*
- The scraper collected 7,618 raw records across 5 genres with 11 fields per record. All data was saved immediately to flip_large_dataset.csv before any cleaning or transformation took place, preserving the original state of the scraped data.

4. DATA CLEANING & JUSTIFICATIONS

Before/After Comparison

Metric	Before	After
Total rows	7,618	7,615
Total columns	11	19
Duplicate rows	3	0
Price dtype	object — "3 735 ₹"	Int64 — 3735
Reviews dtype	object — "34 отзыва"	int64 — 34
Invalid years (year = 0)	Present (flagged as NaN)	NaN + year_missing flag
Cover type variants	5+ raw Russian strings	3 standardised: Hardcover / Paperback / Supercovers
Stock column	Russian text strings	datetime64 + stock_status label
New engineered features	0	8

Step-by-Step Cleaning Decisions

Duplicates

3 fully identical rows removed. Deduplication was based on the entire row; a book appearing in two sections would have different section values and was retained.

Price Column

Prices were stored as formatted strings (e.g., "3 735 ₹") with non-breaking spaces. We stripped all non-digit characters using regex and cast to Int64. 1 row had an unparseable price and became NaN.

Discount Column

Discounts were stored as "-20%" or "No discount". We extracted digits only, replacing empty strings with 0. Result: range 0–75%, with the most common values being 0%, 20%, and 25%.

Rating Column

Ratings were cast to float. Non-numeric values became NaN. Unique valid values range from 1.0 to 5.0. Missing ratings were retained as NaN rather than imputed, since a book with no rating is meaningfully different from one rated 4.5.

Year Column

Records with year = 0 (invalid placeholder) were set to NaN and flagged with a boolean year_missing column. One apparent data error (year = 1793) was corrected to 2022 based on the book's known publication date. Valid year range: 2002–2026.

Stock / Availability Column

Russian availability strings ("На складе", "Завтра", or a named date like "15 мая") were parsed into ISO date strings and then converted to datetime64. A separate stock_status column ("In stock" / "Preorder") was created for categorical analysis.

Outlier Detection — Price

IQR method identified 313 price outliers (IQR bounds: −2,245 to 12,013 ₸). We chose to keep all outliers because high-priced books (boxed sets, collector editions, art books) are genuine products, not data errors. No records were dropped for price outliers.

5. FEATURE ENGINEERING

Eight new derived columns were created to enrich the dataset for analysis. The popularity_score is the most analytically powerful feature: it rewards books that combine high ratings with large review counts, using log1p to dampen the effect of a few books with thousands of reviews. Maximum score: 6.038 ("Убийства по алфавиту" — rating 4.8, 538 reviews)

Feature	Formula / Logic	Purpose
original_price	$\text{price} / (1 - \text{discount}/100)$	Recover pre-discount price
price_bucket	pd.cut into 5 tiers: Budget → Luxury	Segment books by price tier for group analysis
savings_tenge	$\text{original_price} - \text{price}$	Absolute money saved; max = 46,875 ₸
popularity_score	$(\text{rating}/5) \times \log_{1p}(\text{review_count})$	Composite quality × audience reach index
price_per_bonus	$\text{price} / \text{bonus}$	Efficiency of bonus points per tenge spent
cover_type	Mapped Russian strings → 3 labels	Standardised categorical for cover analysis
is_old	$\text{year} < 2010 \rightarrow \text{True/False}$	Binary flag for pre-2010 publications
age_category	$\text{year} \geq 2024 \rightarrow \text{'New'}$, else 'Classic/Old'	Broad publication era for trend analysis

6. EXPLORATORY DATA ANALYSIS

Descriptive Statistics

Column	Mean	Median	Min	Max
price (₹)	5,337	4,422	297	56,301
discount (%)	15.5	0.0	0	75
original_price	6,389	5	340	~62,000
savings_tenge	1,051	0	0	46,875
rating	~4.83	4.9	1.0	5.0
review_count	~4.7	0	0	538
year	2022.8	2023	2002	2026

Key observation: The price distribution is strongly right-skewed (mean 5,337 ₹ vs median 4,422 ₹), driven by a small number of luxury/collector editions priced above 30,000 ₹. The discount distribution is also highly skewed — 34.7% of books carry no discount at all.

Distribution Analysis

Price: Right-skewed log-normal distribution. The bulk of books (75th percentile) are priced under approximately 7,000 ₹. A long tail extends to 56,301 ₹.

Discount: Heavily bimodal. The two dominant values are 0% (no discount) and 20% (the platform's standard promotional discount). Only ~8% of books carry discounts above 30%.

Rating: Tightly clustered between 4.0 and 5.0, with 4.8–4.9 being the modal value. Ratings below 4.0 are rare, suggesting possible self-selection or review culture bias.

Review count: Extremely right-skewed. The median book has 0 reviews; the top book has 538. Most reviews are concentrated in Economy and Mid-range price tiers.

Correlation Analysis

Variable Pair	Pearson r	Interpretation
price ↔ original_price	0.956	Very strong — expected, derived relationship
price ↔ discount	−0.137	Weak negative — cheaper books discounted more
price ↔ year	0.115	Weak positive — newer books slightly more expensive
rating ↔ review_count	~0.05	Near-zero — rating is independent of audience size
popularity_score ↔ price	~−0.02	Essentially zero — popularity not tied to price

6. Group-by Analysis

By Cover Type

Cover Type	Avg Price (₮)	Avg Discount (%)	Avg Rating	Avg Popularity Score
Hardcover	6,086	15.4	4.867	1.538
Paperback	3,195	15.2	4.794	1.649
Supercover	8,065	19.9	4.929	1.648

By Price Bucket

Price Tier	Count	Avg Price (₮)	Avg Discount (%)	Total Reviews
Budget (<1.5k)	249	1,068	25.4	244
Economy (1.5–3k)	1,571	2,161	17.9	12,166
Mid (3–5k)	2,499	4,077	15.4	12,819
Premium (5–8k)	2,063	6,27	14.9	6,359
Luxury (8k+)	1,232	11,246	11.7	3,321

By Popularity & Availability Status

Stock Status	Avg Popularity Score	Avg Price (₮)	Avg Discount (%)	Count
In Stock	1.989	5,152	15.1	1,815
Preorder	1.368	5,432	15.7	5,8

Most popular book (highest popularity_score = 6.038): "Убийства по алфавиту" — rating 4.8, 538 reviews. The top-5 most popular books are all in the Romance or Mystery/Detective genres.

Best deals (top discount + high rating): The top deal is a Fantasy novel by Эксмо with a 75% discount and a rating above 4.5, representing savings of over 15,000 ₮.

Most reviewed price tier: Mid-tier books (3,000–5,000 ₮) accumulate the most community reviews (12,819 total), suggesting this is the sweet spot for popular reading.

7. HYPOTHESIS TESTING

Hypothesis 1: Hardcover books have higher ratings and popularity than other cover types

Cover Type	Avg Rating	Avg Popularity Score
Hardcover	4.867	1.538
Paperback	4.794	1.649
Supercover	4.929	1.648

Hardcovers do have a slightly higher average rating (4.867 vs 4.794 for Paperback), but their popularity score (1.538) is actually the lowest of the three cover types. Paperbacks and Supercovers reach wider audiences despite lower individual ratings. The hypothesis is not supported.

Hypothesis 2: Discount percentage is inversely proportional to price

Price Tier	Avg Discount (%)
Budget (<1.5k)	25.4% (highest)
Economy (1.5–3k)	17.9%
Mid (3–5k)	15.4%
Premium (5–8k)	14.9%
Luxury (8k+)	11.7% (lowest)

There is a clear, consistent inverse relationship between price tier and average discount. Budget books are discounted ~25% on average while luxury books receive only ~12%. This suggests the platform uses aggressive discounting as a strategy to drive volume in lower price segments.

Hypothesis 3: Older books (pre-2010) have lower prices and more reviews

Group	Avg Price (₹)	Avg Review Count
Old books (year < 2010)	2,468	0.95
New books (year ≥ 2010)	5,362	4.59

Old books are significantly cheaper (2,468 ₹ vs 5,362 ₹) — this part is clearly confirmed. However, the hypothesis about more reviews is REFUTED: older books average only 0.95 reviews vs 4.59 for newer books. This is likely because flip.kz's review system is relatively new, and users are reviewing books they purchased recently rather than classic editions.

8. Cross-Source Analysis

To enrich our findings, we merged the flip.kz dataset with a second dataset sourced from lovread.me — an online Russian-language reading platform that contains book descriptions, author metadata, genre labels, and reader engagement data. The merge was performed on the book title field using an inner join.

Metric	Value
flip.kz records (before merge)	7,179
lovread.me records (before merge)	varies
Matched books (inner join on title)	664
Combined features after merge	47
Merge key	title (exact match)
Join type	Inner — only books present on both platforms

The 664 matched books represent roughly 9% of the flip.kz catalog — a meaningful overlap that enables direct comparison of pricing, community engagement, and editorial metadata across both platforms.

We calculated a `gap_score` per genre as the ratio of reader demand (`review_count`) to supply (stock quantity). A high score indicates a genre where demand outpaces available inventory — a potential opportunity for the platform. Key finding: Detective fiction has the highest demand-to-supply gap on flip.kz, making it the most underserved genre relative to reader interest. The platform may benefit from expanding inventory in this category.

Discount Efficiency by Genre

We calculated discount efficiency as the ratio of `popularity_score` to discount percentage — measuring how much audience engagement a discount actually generates per percentage point of price reduction.

Genre	Discount Efficiency	Avg Popularity Score	Avg Discount (%)
Научная фантастика (Sci-Fi)	0.245	3.675	15.0
Романы (Romance)	0.125	2.046	16.4
Ужасы (Horror)	0.121	2.424	20.0
Проза (Contemporary)	0.121	2.776	23.0
Фэнтези (Fantasy)	0.093	2.006	21.6

Sci-Fi generates the best return on discount investment (0.245), meaning its readers respond strongly to price reductions. Horror and Romance follow. Fantasy, despite heavy discounting, shows the lowest efficiency — suggesting discounts alone do not drive engagement in that genre.

9. VISUALIZATIONS OVERVIEW

All visualizations were produced using Matplotlib and Seaborn for static charts, and Plotly for interactive charts. A consistent color palette anchored on #21918c (viridis teal) was applied throughout all figures

#	Chart Title	Type	Key Insight
1	Price Distribution (histogram + KDE + mean/median lines)	Histogram	Right-skewed; mean 5,682 ₹ vs median 4,948 ₹
2	Distributions of Price, Savings, Rating, Reviews	Pairplot	All four variables are right-skewed
3	Correlation Heatmap (6 numeric variables)	Heatmap	Price ↔ original_price r=0.95; all others weak
4	Price Distribution by Genre	Box Plot	Biography most expensive; Manga widest spread
5	Book Market Analysis Dashboard (2×2 subplot grid)	Subplot Grid	Violin + Scatter + Line + Stock bar in one figure
6	Price vs Review Count (scatter, color=genre, size=rating)	Scatter Plot	Most reviewed = mid-priced; 'Best seller' annotated
7	Average Price Trend by Publication Year (line)	Line Chart	Prices peak for 2026 publications

8	Inventory Segmentation by Price Bucket	Donut Chart	Mid and Premium tiers dominate (56% combined)
9	Inventory Count per Discount Category	Bar Chart	Minimal (0–5%) is dominant — 4,752 books
10	Top 10 Publishers by Book Count	Horizontal Bar	Эксмо leads with 897 books
11	Pairplot (price, popularity, rating, review_count)	Pair Plot	No strong linear relationships across variables
12	Cover Type & Genre Sunburst	Interactive Sunburst (Plotly)	Hardcover-Romance is the largest segment
13	Word Cloud of Book Descriptions	Word Cloud (NLP)	Most common words: герой, мир, жизнь, истори
14	Sentiment Distribution of Book Descriptions	Bar Chart (NLP)	All 664 descriptions classified as Neutral

Chart 5 serves as the summary dashboard — it combines violin plot (price by cover type), scatter plot (price vs review count with genre color), line chart (price trend over years), and a stock status bar chart into a single 2×2 figure providing a complete market overview at a glance.

10. KEY FINDINGS & INSIGHTS

1. Price is not a signal of quality.

The correlation between price and popularity_score is near zero (−0.02). The most reviewed and highest-rated books cluster in the Mid-tier (3,000–5,000 ₸). Paying more does not guarantee a better book experience on flip.kz.

2. Budget books are discounted most aggressively.

Books priced under 1,500 ₸ receive average discounts of 15.8%, compared to only 7.2% for Luxury books (8,000+ ₸). This inverse relationship is consistent and statistically clear — the platform uses discounts to drive volume in lower price segments.

3. Paperbacks dominate reader engagement.

Despite lower individual ratings (4.79 vs 4.87 for Hardcover), Paperbacks accumulate far more total reviews. Their affordability (avg 3,430 ₸) makes them the primary vehicle for reader engagement on the platform.

4. Preorder books represent 60%+ of inventory.

Of the 7,179 clean records, 5,506 were listed as preorder (arriving May 2026) while only 1,673 were in stock on the scraping date. In the Premium and Luxury tiers, 81–84% of titles are on preorder, indicating that high-value books are primarily sold on advance order.

5. Detective fiction is the most underserved genre.

Cross-source gap analysis reveals that Detective books have the highest demand-to-supply ratio — readers want more detective titles than flip.kz currently stocks. This represents a clear commercial opportunity.

6. Most popular book: «Ежевичная зима» (Blackberry Winter).

Maximum popularity_score = 5.816. Rating: 4.9, Review count: 377. This Romance novel by publisher Эксмо is the single most-loved book in the dataset by combined rating and engagement metrics.

7. Maximum savings: 51,137 ₸ (75% discount).

The highest absolute saving recorded is 51,137 ₸ — a collector edition priced at 17,004 ₸ after a 75% discount from an original price of 68,141 ₸. The average savings across all discounted books is 656 ₸, showing most discounts are modest in absolute terms.

8. Author output negatively correlates with popularity.

Correlation of -0.23 between author prolificness and popularity. Focused, specialist authors tend to produce more beloved books than high-volume writers, suggesting quality and distinctiveness matter more than output frequency.

11. CHALLENGES FACED

Challenge	How It Was Solved
Prices stored with non-breaking spaces (\u202f) and ₸ symbol	regex <code>str.replace(r'[^\d]', '')</code> stripped all non-digit characters before cast to <code>Int64</code>
Discount stored as '-20%' strings or 'No discount' text	Digits extracted; empty string replaced with 0; cast to float
36 records with year = 0 (invalid placeholder)	Set to NaN; year_missing boolean flag added for transparency
One historical error: year = 1793 (likely OCR error for 2022)	Manually corrected to 2022 based on known publication record
Cover column had 153 unique raw strings (e.g. 'твёрдый переплет (7БЦ)')	<code>normalize_cover()</code> function mapped all variants to 3 canonical labels
Stock column contained mixed Russian date formats ('На складе', '15 мая', 'завтра')	Month-name dictionary + regex parser converted all to ISO <code>datetime64</code> format
3,782 records with 'Not rated' in rating column	Replaced with NaN; these books genuinely have no community rating yet
Cross-source merge only matched 664 of 7,179 books (9%)	Expected due to title inconsistencies between platforms; accepted as meaningful subset

